llmXive Automated Review of OmniDirector: General Multi-Shot Camera Cloning without Cross-Paired Data

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of citations within the manuscript.

Metric Definition Error (Section 5.1): The definition of the metric T-Pre contains a fundamental physical inconsistency. The text states: "T-Pre denotes the proportion of predictions with a relative translation error below 20°." Translation error (RTE) measures displacement in spatial units (e.g., meters or normalized scene units), whereas degrees (°) are units of angular measurement used for rotation (RRE). Claiming a translation error threshold in degrees is physically nonsensical. This suggests either a severe typo (confusing the threshold for RRE with RTE) or a misunderstanding of the metric itself. Consequently, the quantitative results in Table 1 (e.g., T-Pre = 72.74%) are currently uninterpretable and cannot be verified against the stated definition.

Unsupported Causal Claim (Section 5.2): In the quantitative comparison, the authors assert that LTX-LoRA's ability to generate shot transitions is "actually an artifact of information leakage rather than genuine camera control." While Table 1 shows LTX-LoRA has a high leakage rate (15.04%), high leakage does not logically necessitate that the transition capability is an *artifact* of that leakage. It is possible for a model to learn transition patterns while also leaking content. The paper provides no ablation study or qualitative evidence isolating the transition mechanism from the leakage to support this specific causal attribution. This claim overstates the evidence provided by the leakage metric alone.

Citation Validity (Section 5.1 & 5.2): The manuscript repeatedly cites and relies on Qwen3-VL (citing `bai2025qwen3`) for the Prompt Expansion Agent. The citation refers to a 2025 preprint. Given the current date, Qwen3 is not a publicly established model version (Qwen2.5 is the current state-of-the-art). If this paper is intended for a future venue (2026), the specific capabilities attributed to Qwen3-VL are speculative. If the model does not exist or the citation is a placeholder, the description of the "Prompt Expansion Agent" methodology is factually unsupported. The authors must clarify the existence and version of the model used.

Citation Consistency: The paper cites `lin2025dpa3` (Depth Anything 3) for camera pose estimation. While the citation exists in the bibliography, the claim that this specific model is used for "robust" estimation in complex real-world scenes requires verification that the cited model actually outperforms existing baselines (like DPT or Depth Anything V2) in the specific context of video pose estimation, which is not explicitly detailed in the text. However, the primary issue remains the metric definition and the causal claim regarding leakage.

Action items.

- [science] Claim that LTX-LoRA's transitions are 'artifacts of leakage' lacks evidence. High leakage does not prove the transition mechanism is an artifact; ablation or qualitative proof is needed to support this causal assertion.
- [writing] T-Pre definition uses degrees (°) for translation error, which is physically incorrect. Translation is a distance metric, not angular. This typo or error makes the

reported 72.74% metric uninterpretable.

- [writing] The paper relies on 'Qwen3-VL' (citing a 2025 preprint) which may not exist yet. The methodology's validity depends on this model's actual capabilities. Clarify if this is a hypothetical citation or a real, available model.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The top 3D visualization is clear with consistent color coding and legends, but the caption fails to describe the four-row bottom section (reference video, camera grid, and two synthetic outputs), creating a mismatch between the figure content and its explanation.

Figure 2

The figure effectively visualizes the grid distortions, but the bottom row contradicts the caption's description of a dolly zoom by showing the subject shrinking rather than remaining fixed in size. Additionally, the text annotations within the bottom panels are too small to read.

Figure 3

Figure 3 provides a clear and comprehensive overview of the OmniDirector architecture, but the caption uses the term 'PE Agent' which contradicts the 'Camera Prompt Generator' label in the diagram, and the color coding in the visual encoding section lacks an explicit legend.

Figure 4

Figure 4 is a clear and well-constructed donut chart that effectively visualizes the evaluation set distribution. All categories are explicitly labeled with percentages on the chart and defined in the legend, and the total sample size is provided.

Figure 5

The figure provides a clear visual comparison of camera cloning results across different methods, but the caption contains a spelling error ('Qualitive') and the method label 'LTX-LoRA' appears potentially misspelled.

Figure 6

Figure 6 presents a visual ablation study but lacks clear definitions for the abbreviated method names ('Tran. PE', 'Sem. Fusion') in the caption, hindering the reader's ability to understand the specific components being removed. Additionally, the connection between the reference video shots and the generated outputs is not explicitly detailed.

Figure 7

The figure effectively demonstrates the visual results of using Canny edges and raw video as conditions, but the top section's labeling is inconsistent with the bottom section, and the 'Synth' column in the top section displays static images rather than motion sequences.
Action items.

- [science] Figure 1: The caption states 'Orthogonal lines represent the ceiling and floor (red and blue)', but the Top View plot shows a grid of red and blue lines on the X-Z plane (floor/ceiling) while the Front/Side views show yellow vertical lines (walls). However, the Front/Side view axes are labeled 'Y Axis' (vertical) and 'X/Z Axis' (horizontal), but the grid lines are yellow. The caption says vertical lines (yellow) denote walls, which matches the Front/Side views. But the Top View shows red and blu
- [science] Figure 2: The 'Dolly Zoom' row shows the subject (green cube) shrinking significantly across the three panels, contradicting the caption's claim that the subject 'remains fixed in size'.
- [writing] Figure 2: The text labels inside the bottom row panels (e.g., 'TRACKING', 'CUBE=3D subject') are illegible due to low resolution.
- [writing] Figure 3: The caption refers to a 'PE Agent' in the bottom panel, but the diagram labels this component as 'Camera Prompt Generator', creating a terminology mismatch.
- [writing] Figure 3: The 'Visual Encoding' section uses color-coded blocks (blue, pink, green) for latent variables, but lacks a legend explicitly mapping these colors to the specific inputs (Reference Image, Camera Grid, Noisy Latent).
- [writing] Figure 5: The caption contains a spelling error ('Qualitive' instead of 'Qualitative').
- [writing] Figure 5: The label 'LTX-LoRA' is likely a typo for 'LTXV-LoRA' or similar, given the context of video generation models, though this cannot be confirmed without external knowledge.
- [science] Figure 6: The row labels 'w/o Tran. PE' and 'w/o Sem. Fusion' are ambiguous;

the caption does not define what 'Tran. PE' or 'Sem. Fusion' stand for, making the ablation study's specific contributions unclear.

- [science] Figure 6: The 'Ref. Video' rows are labeled but the corresponding generated results in the 'Full' rows are not explicitly linked to specific reference shots (e.g., 'first-person view', 'medium shot'), making it difficult to assess if the model correctly cloned the specific camera motion.
- [writing] Figure 7: The top section is labeled 'Reference Video Condition' but lacks a column header for the 'Ref' input, unlike the bottom section which explicitly labels 'Ref', 'Condition', and 'Synth'.
- [writing] Figure 7: The top section's 'Synth' column displays static images (e.g., Harry Potter, deer) rather than video frames or motion sequences, which contradicts the caption's claim of driving 'camera motion'.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not consistently defined, creating barriers for non-specialist readers.

First, the acronym MMDiT (Multi-Modal Diffusion Transformer) is used in the Abstract and Introduction without being spelled out. While common in the field, a general audience requires the full term at first mention. Similarly, 6DoF (six degrees of freedom) appears in the Related Work section without expansion.

The term PE Agent (Prompt Expansion Agent) is introduced in the Method section as a "hierarchical Prompt Expansion (PE) Agent," but the abbreviation is then used repeatedly (e.g., "w/o Trans PE") without a clear, standalone definition of what "PE" stands for in the context of the agent's function. The text should explicitly state "Prompt Expansion (PE) Agent" and ensure the acronym is only used thereafter if necessary, or simply use the full name for clarity.

In the Experiments section, LoRA is used without definition. This should be expanded to "Low-Rank Adaptation" on first use. Furthermore, the metrics T-Pre and R-Pre are introduced as "Translation Precision" and "Rotation Precision" in the text but are immediately referred to by these non-standard abbreviations. The paper should either define these clearly as "Translation Precision (T-Pre)" or, preferably, use the plain English terms throughout to avoid confusion with standard statistical terms.

The metric GSB (Good/Same/Bad) is mentioned in the Experiments section without prior definition. The text should explicitly state "Good/Same/Bad (GSB) comparison" before using the acronym.

Finally, specific model names like TransNet-V2 and DPA-V3 are cited without briefly explaining their function (shot transition detection and camera pose estimation, respectively) for readers who may not be familiar with these specific tools. While the citations exist, a brief parenthetical explanation would improve accessibility.

These changes are minor but essential for ensuring the paper is accessible to a broader scientific audience beyond immediate specialists in video diffusion models.

Action items.

- [writing] Define 'MMDiT' at first use in the Introduction. The acronym appears in the abstract and introduction without expansion, hindering non-specialist readers."
- [writing] Replace '6DoF' with 'six degrees of freedom' on first occurrence in Related Work to ensure accessibility for readers unfamiliar with robotics/graphics terminology."
- [writing] Define 'PE Agent' or 'Prompt Expansion Agent' fully before using the abbreviation. The term 'PE Agent' appears in the Method section without prior definition."
- [writing] Replace 'LoRA' with 'Low-Rank Adaptation' at first mention in the Experiments section. Acronyms for specific techniques should be defined for general audiences."
- [writing] Define 'GSB' (Good/Same/Bad) explicitly before using the abbreviation in the Experiments section. The metric name is introduced as an acronym without context."
- [writing] Replace 'T-Pre' and 'R-Pre' with 'Translation Precision' and 'Rotation Precision' (or similar plain terms) on first use, or define them immediately. These are non-standard abbreviations that obscure meaning."
- [writing] Define 'TransNet-V2' and 'DPA-V3' upon first mention in the Method section. While these are model names, their function (shot detection, pose estimation) should be briefly clarified for non-experts."

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent methodology for camera cloning using a "camera grid" representation. The logical flow from the problem (data scarcity in cross-paired methods) to the solution (visualizing camera motion in an empty 3D grid) is sound. The architectural choices (token concatenation, MMDiT) follow logically from the proposed representation.

However, there are significant logical gaps in the definition and justification of the evaluation metrics, which undermines the validity of the quantitative claims:

1. Metric Definition Inconsistency (Section 4.1.1): The paper defines Relative Translation Error (RTE) and the corresponding precision metric (T-Pre) using angular thresholds

(20 degrees). Translation is a vector quantity with magnitude and direction. While one can measure the angular error of a translation vector, standard RTE in computer vision is a Euclidean distance error (e.g., in meters). If the authors are measuring the angle of the translation vector, this is a non-standard definition that must be explicitly distinguished from distance-based error. The current text conflates "directional error" with a generic "translation error" without clarifying that magnitude is ignored, which is a critical logical flaw in the metric's definition.

2. Evaluation Ground Truth Ambiguity (Section 4.1.1): The authors justify using LLM-based GSB metrics by stating that pose estimation methods (like DPA-V3) suffer from inherent errors in complex scenes. However, the primary quantitative results (Table 1) rely on DPA-V3 to extract poses from *both* reference and generated videos to compute RRE and RTE. If the "ground truth" poses are also derived from an estimator with "inherent errors," the reported RRE/RTE metrics measure the consistency between two noisy estimators, not the true geometric accuracy of the generation. The paper fails to logically distinguish between "estimator noise" and "generation error," making the quantitative superiority claims difficult to validate without a known ground truth or a more rigorous error analysis of the estimator itself.
3. Causal Attribution of Failure (Section 4.2.2): The authors attribute LTX-LoRA's ability to handle shot transitions solely to "information leakage," citing its high leakage rate as proof. This is a logical non-sequitur. A model can simultaneously leak content (high leakage) and learn temporal transitions (high transition accuracy). The presence of leakage does not logically prove that the transition capability is an artifact; it merely indicates a lack of disentanglement. The claim that the transitions are *not* genuine learning requires evidence that the transitions fail when leakage is controlled for, or a mechanistic explanation of why leakage *necessitates* the transition behavior, which is currently missing.

Action items.

- [science] Section 4.1.1 defines RTE and T-Pre using angular thresholds (20 degrees) for translation error. Translation is typically a distance metric, not an angle. Explicitly clarify if this measures the angular deviation of the translation vector to avoid confusion with standard Euclidean distance errors.
- [science] Section 4.1.1 claims pose estimators have inherent errors, yet Table 1 relies on DPA-V3 for both reference and generated poses to compute RRE/RTE. If ground truth is also estimated, metrics measure estimator consistency, not generation accuracy. Clarify if true ground truth exists or if the 'errors' refer to the evaluation noise floor.
- [science] Section 4.2.2 claims LTX-LoRA's transition success is an 'artifact of leakage' due to high leakage rates. High leakage does not logically preclude learning transitions. Provide evidence that transitions are side-effects of leakage rather than genuine learning, as the current causal link is unsupported.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the scope and capabilities of OmniDirector that exceed the evidence provided in the experiments.

First, the term "director-level control" is used repeatedly in the Abstract, Introduction, and Conclusion to describe the framework's capabilities. While the paper successfully demonstrates high-fidelity cloning of camera trajectories and shot transitions, the term "director-level" implies control over narrative structure, emotional tone, and character performance. The current evaluation metrics (RRE, RTE, leakage rate) only measure geometric and content fidelity, not narrative or emotional alignment. Without specific experiments or metrics addressing these higher-level cinematic attributes, the claim of "director-level control" is an overreach. The authors should qualify this claim to reflect the specific scope of camera motion and shot transition control demonstrated.

Second, the "Emergent Camera Understanding" section (Section 5.3) presents a significant over-interpretation of the results. The authors claim that the model can clone complex effects like the "Hitchcock zoom" (dolly zoom) and fisheye distortion "without requiring explicitly curated training data for such specific trajectories" (lines 430-435). However, Figure 2 explicitly shows that the training data generation process includes rendering dolly zooms and fisheye distortions. Therefore, the model's ability to reproduce these effects is a result of exposure to these specific patterns during training, not an emergent zero-shot capability. The claim of "emergent" behavior is misleading and contradicts the methodology described in Section 3.1.2. The text should be revised to accurately reflect that these capabilities are learned from the curated training distribution.

Finally, the claim that the Prompt Expansion agent "completely decouples" the camera signal from the reference video (Section 4.2, lines 330-332) is too absolute. The quantitative results in Table 1 show a non-zero leakage rate of 3.38% for the proposed method. While this is a significant improvement over baselines, it is not "complete" decoupling. The language should be adjusted to "significantly reduces" or "effectively decouples" to maintain scientific rigor and align with the reported metrics.

Action items.

- [writing] The claim of 'director-level control' (Abstract, Intro) is overreaching. The paper demonstrates camera motion cloning but lacks evidence for controlling narrative depth or emotional atmosphere. Qualify this claim to 'camera motion control' or provide metrics for narrative alignment.
- [science] The 'Emergent Camera Understanding' section (Section 5.3) overstates generalization. Claiming the model executes dolly zooms 'without requiring explicitly curated training data' (lines 430-435) is unsupported, as Figure 2 shows these were rendered in

training. Clarify that this is learned behavior, not zero-shot emergence.

- [writing] The assertion that the Prompt Expansion Agent 'completely decouples' camera signals (Section 4.2, lines 330-332) is absolute and unsupported. Table 1 shows a 3.38% leakage rate. Replace 'completely decouples' with 'significantly reduces leakage' to align with quantitative evidence.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel framework for camera motion cloning but lacks sufficient disclosure regarding data privacy, third-party API usage, and potential dual-use risks.

Data Privacy and Third-Party APIs: The methodology relies heavily on external Large Multimodal Models (LMMs) for critical tasks. Specifically, Section 3.2.1 describes using Qwen3-VL to refine camera prompts, and Section 4.1 utilizes Gemini 3.1 Pro for quantitative evaluation (GSB comparisons). The paper does not state whether these API calls involve transmitting the reference video frames or generated content to external servers. Given that the input videos are sourced from the internet (Section 4.1), there is a risk of inadvertently processing copyrighted material, PII, or sensitive content through these third-party services. The authors must explicitly disclose the data handling policies of these APIs and confirm that no sensitive data is retained or used for further training by the API providers.

Dataset Ethics and Copyright: The training set consists of 1.8M internet videos (Section 4.1). The manuscript does not mention any Institutional Review Board (IRB) approval, consent mechanisms, or copyright clearance procedures. In the context of generative AI, the use of uncurated internet-scale datasets raises significant ethical concerns regarding the rights of content creators and the potential inclusion of non-consensual or harmful content. A statement clarifying the legal basis for data usage, the filtering of PII, and adherence to copyright laws is required.

Dual-Use and Misinformation Risks: The "emergent" capability described in Section 4.3, where the model can clone camera motion from raw RGB videos without explicit parameter extraction, significantly lowers the technical barrier for generating realistic video manipulations. This capability could be misused for creating deepfakes, spreading misinformation, or generating non-consensual content. The paper currently lacks a discussion on these dual-use risks. The authors should add a dedicated "Ethical Considerations" section outlining the potential for misuse and any safeguards implemented (e.g., watermarking, usage restrictions) to mitigate these risks.

Action items.

- [writing] The manuscript relies on Qwen3-VL and Gemini 3.1 Pro for prompt generation and evaluation (Sections 3.2.1, 4.1). Explicitly disclose the data privacy policies of these third-party APIs regarding the reference videos and generated prompts to ensure no sensitive content is inadvertently transmitted or stored.
- [writing] The training dataset comprises 1.8M internet videos (Section 4.1) without mentioning IRB approval, consent procedures, or copyright compliance. Add a statement clarifying the legal and ethical basis for using this data, specifically addressing potential copyright infringement and the exclusion of personally identifiable information (PII).
- [writing] The "emergent" capability to clone camera motion from raw RGB videos (Section 4.3) lowers the barrier for generating deepfakes or misleading content. Include a dedicated "Ethical Considerations" subsection discussing potential dual-use risks (e.g., misinformation, non-consensual content) and proposed mitigation strategies.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence for OmniDirector is conceptually strong but lacks the statistical rigor required to fully support its quantitative claims of superiority.

The primary concern is the ground-truth definition for the core metrics. The authors use DPA-V3 (Section 5.1) to extract camera poses from reference videos to calculate Relative Rotation Error (RRE) and Relative Translation Error (RTE). DPA-V3 is a monocular depth/pose estimator, which is inherently susceptible to scale ambiguity and temporal drift, particularly in the "complex real-world scenes" the authors claim to evaluate. The paper asserts "scale-invariant" metrics but does not demonstrate that the ground-truth labels themselves are robust to the specific failure modes of DPA-V3. If the ground truth has a high noise floor, the reported improvement of OmniDirector (RRE 2.64 vs. 4.11 for CamCloneMaster) may be an artifact of the model learning the specific biases of the DPA-V3 estimator rather than true geometric fidelity. A sensitivity analysis or a comparison with a multi-view ground truth (where available) is necessary to validate these numbers.

Secondly, the reliance on Gemini 3.1 Pro for the GSB (Good/Same/Bad) evaluation (Table 3) and semantic transition accuracy (Sem-Pre) introduces a significant validity threat. The paper provides no details on the prompts used, the temperature settings, or, crucially, any validation of the LLM's scoring against human raters. In generative AI research, LLM-as-a-judge metrics are notoriously prone to bias, especially when the judge model (Gemini) and the generator (OmniDirector) may share training data or when the judge is evaluating its own domain of expertise. The claim that OmniDirector achieves 94.26% narrative consistency based solely on an LLM score is not robust without human verification or a clear demonstration of the judge's correlation with human preference.

Finally, the ablation study in Table 2 presents dramatic performance drops (e.g., Sem-Pre falling from 83.79% to 38.45% without Trans PE) without reporting statistical significance. While the effect size is large, the absence of standard deviations, confidence intervals, or a description of the random seeds used for these specific ablation runs makes it difficult to rule out variance as a factor. Given the high stakes of these claims for the proposed "hierarchical prompt expansion" mechanism, the evidence needs to be more statistically grounded.

The paper is sound in its conceptual approach, but the quantitative evidence requires strengthening through more robust ground-truth validation and human-in-the-loop evaluation to rule out evaluator bias.

Action items.

- [science] Ground truth for RRE/RTE relies on DPA-V3, a monocular estimator prone to scale drift. The paper lacks a sensitivity analysis or error bounds for these labels. Without quantifying the evaluator's noise floor, reported improvements may reflect estimator bias rather than true model superiority.
- [science] GSB and Sem-Pre metrics rely solely on Gemini 3.1 Pro without human validation or inter-rater reliability checks. Relying on an LLM judge for primary qualitative benchmarks introduces significant bias risk. Please provide human evaluation results or validate the judge's correlation with human preference.
- [science] Ablation study (Table 2) shows large metric drops (e.g., Sem-Pre 83% to 38%) but lacks statistical significance reporting. Please provide standard deviations, confidence intervals, or details on random seeds to rule out variance as a factor in these dramatic performance changes.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the "Experiments" section requires clarification and additional rigor to support the paper's claims.

First, there is a critical ambiguity in the definition of the T-Pre metric (Section "Evaluation Metrics"). The text states: "T-Pre denotes the proportion of predictions with a relative translation error below 20°." However, translation error (RTE) is a measure of distance (e.g., meters or normalized units), not an angle. Rotation error (RRE) is correctly measured in degrees. It is highly likely that the authors intended to use a distance threshold (e.g., 0.2m or a normalized scale) or a different metric entirely. If the threshold is indeed 20 degrees, the metric is ill-defined for translation. This unit inconsistency must be corrected to ensure the validity of the quantitative comparison in Table 1.

Second, the GSB pairwise comparison results presented in Table 3 (Section "Comparisons with State-of-the-Art Methods") lack statistical validation. The table reports percentages (e.g., 88.52% for Camera accuracy) but does not provide confidence intervals or p-values from a significance test (such as a binomial test or McNemar's test). Without these, it is impossible to determine if the observed superiority over CamCloneMaster is statistically significant or merely a result of sampling variance, especially given the subjective nature of human/LLM evaluation.

Third, the ablation study in Table 2 reports only point estimates for metrics like RRE and RTE. In diffusion-based generation, results can vary significantly across runs due to stochasticity. The absence of standard deviations or confidence intervals makes it difficult to assess the stability of the proposed components (e.g., "w/o Semantic Fusion" vs. "Full"). The authors should report the mean and standard deviation over multiple runs or provide confidence intervals to substantiate the claimed improvements.

Finally, the evaluation set size is stated as 1,094 samples. While this is a reasonable number, the paper does not specify if the results are averaged over multiple seeds or if the evaluation was performed on a single split. Reproducibility of the statistical findings would be enhanced by explicitly stating the number of evaluation runs and the method used to aggregate results.

Action items.

- [science] The evaluation metrics R-Pre and T-Pre are defined with inconsistent units: R-Pre uses a 4° angular threshold, while T-Pre uses a 20° threshold for translation error. Translation error (RTE) is typically measured in distance (meters) or normalized units, not degrees. Clarify the definition of T-Pre and ensure the threshold is appropriate for the metric's unit.
- [science] The GSB (Good/Same/Bad) pairwise comparison results in Table 3 lack statistical significance testing (e.g., binomial test or confidence intervals). With sample sizes implied by the percentages, the observed differences (e.g., 88.52% vs. baseline) should be validated to ensure they are not due to random chance.
- [science] The ablation study in Table 2 presents point estimates for metrics (e.g., RRE, RTE) without reporting variance (standard deviation) or confidence intervals. Given the stochastic nature of diffusion models and the evaluation set size (1,094 samples), uncertainty quantification is necessary to assess the robustness of the reported improvements.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a generally high standard of academic writing, with a clear logical flow from the problem statement to the proposed solution and experimental validation. The abstract effectively summarizes the core contributions, and the introduction successfully motivates the need for a new camera representation. The technical descriptions in the Method section are detailed and largely coherent.

However, there are specific areas where clarity and precision can be improved. In Section 3.1.1, the description of the camera grid rendering is slightly verbose; tightening the sentence structure would enhance readability without losing technical detail. More critically, in Section 4.1, the definition of the T-Pre metric uses "degrees" for translation error ("below 20°"), which is dimensionally inconsistent as translation is a spatial distance, not an angle. This likely represents a typo that could confuse readers regarding the evaluation protocol.

Additionally, the phrasing in Section 3.2.1 regarding the "hierarchical prompt expansion agent" is somewhat abstract. The phrase "understanding signal relationships" lacks specificity; clarifying how the agent technically achieves this understanding would strengthen the prose. Finally, in Section 4.2.2, the assertion that LTX-LoRA's performance is an "artifact of information leakage" is a strong scientific claim. While the writing is grammatically correct, the tone could be refined to ensure the argument is presented as a deduction from the data rather than a definitive conclusion, improving the overall persuasiveness of the text. Addressing these points will polish the manuscript to a publication-ready standard.

Action items.

- [writing] In Section 3.1.1, the sentence 'Only 3D grid lines are used to indicate the directions of spatial coordinate axes, thereby clearly presenting the geometric structure of the space and the camera motion trajectory' is slightly wordy. Consider simplifying to 'Only 3D grid lines indicate spatial axes, clearly presenting the geometric structure and camera trajectory.' to improve flow.
- [writing] In Section 3.2.1, the phrase 'harmoniously integrates different control signals by systematically describing camera motion and visual content through understanding signal relationships' is vague and repetitive. Clarify the specific mechanism of 'understanding signal relationships' or rephrase for conciseness.
- [writing] In Section 4.1, the definition of T-Pre states 'relative translation error below 20°'. Since translation is a vector, 'degrees' is likely a typo for 'meters' or a normalized unit. This ambiguity impairs understanding of the metric.
- [writing] In Section 4.2.2, the sentence 'While LTX-LoRA can generate occasional shot transitions, the results in Table 1 indicate that this capability is actually an artifact of information leakage rather than genuine camera control' is a strong claim. Ensure the phrasing is precise and supported by the preceding text to avoid sounding speculative.